

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This task asks you to explain 'The Shrinking Uji Cup' and answer our driving question: Why does doubling the size of a container make it hold eight times as much, and how can we use this pattern to design, build and predict the properties of any similar shape or object? Use evidence from all eight lessons: the uji cups, your measurements, your constructions, the tables you completed and your Driving Question Board. Write your answers in the space under each prompt. Use full sentences, show every step of your working with units, and use the correct words: similar, corresponding sides, corresponding angles, centre of enlargement, linear scale factor k , area scale factor and volume scale factor.

Part 1: What makes shapes similar?

In Lesson 1 you met three uji cups: a small toddler cup, a standard adult mug and a large catering flask, all made by the same manufacturer. (a) Every dimension of the medium cup is twice the matching dimension of the small cup, and every dimension of the large cup is three times that of the small cup. State the linear scale factor k for small to medium and for small to large. (b) Explain what two properties must be true for two figures to be called similar. (c) Describe how you could check, using a ruler and a protractor, that two Kimbo tins are similar.

(a) Small to medium: $k = 2$. Small to large: $k = 3$. (b) Two figures are similar if and only if (1) their corresponding angles are equal and (2) their corresponding sides are in a constant ratio, which is the linear scale factor k . Similarity is about shape, not size. (c) Measure the height and diameter of both tins with a ruler, then divide the larger measurement by the matching smaller one. If every pair gives the same ratio, the sides are in constant ratio. Then use a protractor, or the corner of a sheet of paper, to compare corresponding angles. If the angles are equal and the ratio of every pair of corresponding lengths is the same k , the tins are similar.

Part 2: Enlargement, the centre and the scale factor

(a) When you traced the pentagon ABCDE and its image,

(a) I drew a straight line (ray) through each pair of corresponding vertices (A to A', B to B', C to C'). The single point where all the rays meet is the centre of enlargement O. (b) $k = \text{image length} \div \text{object length}$. When I measured pairs of corresponding sides, I

how did you find the centre of enlargement O? (b) How do you calculate the linear scale factor k from an object and its image, and what did you find about k when you measured several pairs of corresponding sides? (c) A cartographer at the Survey of Kenya reduces a street map of Kisumu using $k = \frac{1}{2}$. Which properties of the map stay the same as the real layout, and which change? Explain how a scale factor between 0 and 1 differs from one greater than 1.	got the same value of k every time, so k is constant for all corresponding lengths in an enlargement. (c) Angles stay the same and the shape is preserved, so the map has the same shape as the real layout. Every length changes: each length on the map is $\frac{1}{2}$ times the matching length in the original. When k is greater than 1 the image is an enlargement, farther from O. When k is between 0 and 1 the image is a reduction, closer to O. The centre O stays fixed.
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Part 3: Why area scales by k^2

The label of the small uji cup needs a certain amount of paper. (a) Using your measurements from the area investigation with squares and rectangles ($k = 2, 3$ and 4), state the rule that links the linear scale factor to the area scale factor. (b) Explain why this rule works, using the length and width of a rectangle. (c) Use the rule to say how many times as much paper the label of the large uji cup ($k = 3$ from the small cup) needs, and how many times as much for the medium cup ($k = 2$). (d) A shamba in Kericho is enlarged on a plan with $k = 4$. The original plot has an area of 50 m^2. Find the area of the image.	(a) Area scale factor = k^2, the square of the linear scale factor. (b) If a rectangle's length and width are both multiplied by k, the new area is $(k \times \text{length}) \times (k \times \text{width}) = k \times k \times \text{length} \times \text{width} = k^2$ times the original area. Area uses two lengths, so k is used twice. (c) Large cup, $k = 3$: the label needs $3^2 = 9$ times as much paper. Medium cup, $k = 2$: it needs $2^2 = 4$ times as much paper. This is why the numbers were surprising: the label does not need 3 times or 2 times as much. (d) Area scale factor = $4^2 = 16$. Area of image = $16 \times 50 \text{ m}^2 = 800 \text{ m}^2$.
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Part 4: Why volume scales by k^3

(a) State the rule that links the linear scale factor k to the	(a) Volume scale factor = k^3. Volume depends on three lengths (for example length, width and height), and each is multiplied by k, so the volume is multiplied by $k \times k \times k = k^3$. (b) When every dimension is doubled, $k = 2$, so the volume becomes $2^3 = 8$ times as
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volume scale factor of similar solids, and explain why it works. (b) Use it to explain why doubling every dimension of a container makes it hold eight times as much. (c) Two similar cylindrical water tanks in the Rift Valley have a linear scale factor of 3. How many times more paint does the larger tank need, and how many times more water does it hold? (d) Mama Wanjiru's small mandazi tin uses 1 000 g of flour for its dough. She buys a similar tin with $k = 2$. How much flour does she need to fill the large tin to the same level, and why is it not 2 000 g?	large. The container holds eight times as much, not two times. (c) Paint covers surface area, so the larger tank needs $3^2 = 9$ times as much paint. Water fills volume, so it holds $3^3 = 27$ times as much water. (d) Volume scale factor = $2^3 = 8$, so she needs $8 \times 1\,000\text{ g} = 8\,000\text{ g}$ of flour. Doubling the lengths multiplies the volume of the tin by 8, not by 2, so 2 000 g would be far too little. Her large tin needs 8 times the dough.
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Part 5: Answer the driving question	
Write a complete, connected answer to the driving question: Why does doubling the size of a container make it hold eight times as much, and how can we use this pattern to design, build and predict the properties of any similar shape or object? In your answer: (1) refer to the uji cups and their scale factors; (2) explain the roles of k, k^2 and k^3 and what each one measures; (3) give ONE real-life example from Kenya of how similarity and enlargement are used, showing which scale factor you would use and why; and (4) say why the original prediction of 'twice as big means twice as much' was wrong.	The three uji cups are similar solids: corresponding angles are equal and every length on the medium cup is 2 times the matching length on the small cup ($k = 2$), and on the large cup 3 times ($k = 3$). One scale factor k describes how all lengths change, but lengths are only one kind of measurement. Area uses two lengths, so areas are multiplied by k^2. The label of the large cup ($k = 3$) needs $3^2 = 9$ times as much paper as the small cup. Volume uses three lengths, so volumes are multiplied by k^3. Doubling a container's dimensions gives $k = 2$ and a volume of $2^3 = 8$ times as much. Tripling gives $3^3 = 27$ times. Our first prediction, that twice as big means twice as much, was wrong because we treated every quantity as a length, but only lengths scale by k. Area scales by k^2 and volume by k^3. We can use this pattern to design and predict. For example, if a Rift Valley water tank is enlarged with $k = 3$, we know before building it that it needs 9 times as much paint (k^2) and will hold 27 times as much water (k^3). A caterer can scale a recipe in the same way: a similar mandazi tin with $k = 2$ needs 8 times the dough, so 1 000 g becomes 8 000 g. This shows how similarity and enlargement are applied in real life, in packaging, building, catering and map-making.

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Accuracy of similarity and scale factor ideas	States both conditions for similarity (equal corresponding angles, constant ratio of corresponding sides) and calculates and interprets k correctly for enlargements and reductions, including the role of the centre of enlargement.	States the similarity conditions and calculates k correctly, with minor gaps in explaining the centre of enlargement or reductions.	Confuses similar with equal or same size, or cannot find or interpret k .
Use of k , k^2 and k^3	Correctly applies k for lengths, k^2 for areas and k^3 for volumes in every question, and explains why each power occurs by counting the dimensions involved.	Applies the correct rule in most questions and gives a partial explanation of why.	Uses k for area or volume, or applies the rules without explanation.
Calculations and communication	Shows every step with correct arithmetic and units (for example 800 m ² , 8 000 g) and uses correct mathematical vocabulary throughout.	Mostly correct arithmetic and units with some steps or terms missing.	Answers have errors or no working, or units are missing.
Connecting evidence to the phenomenon and real life	Links the uji cups, the tank and mandazi examples to the rules, explains why the first prediction failed, and gives a clear Kenyan application with the correct scale factor.	Refers to the uji cups and one application but the link to the rules or to the failed prediction is incomplete.	Gives a general answer with little evidence from the lessons or no real-life application.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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